

Dyllon A. Dunton | Pittsfield, Maine | (207) 313-9017 | duntondyllon@gmail.com

Robotics Software Engineer | Computer & Electrical Engineering

Portfolio: dyllondunton1.github.io

github.com/DyllonDunton1 | linkedin.com/in/dyllon-dunton

Professional Summary

Robotics software engineer with an M.S. in Computer Engineering and dual B.S. degrees in Electrical and Computer Engineering. Experienced in ROS2, C++/Python, embedded control, mapping, sensor integration, and deployable software for physical systems.

Education

University of Maine – Orono, ME

M.S. Computer Engineering, Robotics & AI Focus, GPA: 4.0, May 2026

B.S. Computer Engineering & B.S. Electrical Engineering, GPA: 3.748, May 2024

Robotics Team Officer — Software/Hardware Co-lead, Treasurer, 2022–2024

Technical Skills

Robotics/Embedded: ROS2, C++, Python, Raspberry Pi, Teensy, STM32, motor control, odometry, LiDAR/IMU/camera integration

Software/Tools: Linux, Docker, Git, SQL, MATLAB, FastAPI, PlatformIO, data logging

Hardware: custom PCB design, power distribution, motor drivers, soldering, 3D-printed integration

AI/Modeling: PyTorch, TensorFlow, OpenCV, reduced-order modeling, system identification

Selected Robotics Projects

Charlie SLAMBot — ROS2 LiDAR Mapping Robot

- Built a ROS2 mobile robot for LiDAR mapping, wheel odometry, and autonomous navigation development using Raspberry Pi 4B, Teensy 4.1, LiDAR, camera, encoders, and stepper motors.
- Integrated Teensy-based closed-loop motor control, encoder feedback, runtime tuning, power distribution, and modular 3D-printed/protoboard hardware.

Submersible Drone for Hull Inspection — Senior Capstone

- Developed the submarine-side embedded system for a tethered ROV hull-inspection platform with ROS2/Ethernet communication, Raspberry Pi 4B, STM32 control, and live camera feedback.
- Designed custom PCBs integrating power rails, motor drivers, sensors, lights, propellers, ballast stepper motors, camera gimbal servos, and IMU-assisted roll/depth control.

Relevant Experience

Graduate Research Assistant — UMaine ECE Dept. | Jul 2024 – Present

- Developed predictive modeling software for offshore wind turbine motion using environmental and motion time-series data from simulated operating conditions.
- Designed physics-informed latent-state models and deployable inference/visualization tooling for physical motion prediction.

Engineering Intern — General Electric Gas Power Bangor | Mar 2021 – May 2024

- Programmed embedded sensing, control, and logging systems for high-throughput manufacturing equipment.
- Built SQL-based inspection-line reporting tools to support real-time production decision-making.

Research Assistant — Advanced Structures & Composites Center | Aug 2022 – Dec 2023

- Developed ROS2/Python calibration software for a multi-camera FLIR Boson array used in 3D-print monitoring and failure detection.
- Containerized calibration and perception workflows with Docker for reproducibility.